
Unveiling the Invisible Stage: A Bayesian Inverse Approach to Audience Dynamics and Fairness Design in Competitive Dancing

Summary

In competitive dancing shows, the tension between professional adjudication (judges) and popularity-driven public voting (audience) often leads to controversial eliminations. In this paper, we present a comprehensive framework that reconstructs hidden voting patterns, analyzes the structural conflicts between judges and fans, and proposes a robust Two-Key voting system.

Decoding the Hidden Vote: A Bayesian Inverse Framework. With raw audience votes unobserved, we cast reconstruction as a **Bayesian Inverse Problem**. We sample feasible vote-share vectors consistent with historical eliminations via Hit-and-Run MCMC and recover a time-consistent MAP trajectory with a **Hidden Markov Model (HMM)**. The hybrid pipeline reproduces eliminations with **99.6% accuracy** over 34 seasons.

Analyzing the Mechanics of Controversy. We compare Rank vs. Percent aggregation using two metrics: **Fan-Favor Index (FFI)** and a **Suppression Score**. Although Percent is more fan-aligned on average, it can penalize extreme “high-fan/low-judge” cases by raising the survival threshold near the bottom; a judges-save step acts as a circuit breaker in knife-edge weeks.

Disentangling Influence Factors. Using two-path regressions on 2,777 contestant-weeks, we find structural misalignment ($\rho \approx -0.19$): judges reward **technical form** (coaching), while fans respond to **persona** and visibility. Several effects vary over weeks, helping explain mid-season reversals where early favorites fade.

Designing a Fairer Future: The Two-Key System. To reconcile legitimacy and engagement, we propose **Two-Key + Live Save**: separate judge/fan leaderboards define a danger pool, fans save one contestant live, and judges enforce a technical floor for elimination. Counterfactual simulations place the system on the Pareto frontier, improving **Legitimacy** and **Robustness** while maintaining high **Engagement**.

Keywords: Bayesian Inverse Problem, Feasible Region Sampling, Hidden Markov Model, Voting System Design, Pareto Optimality, Multi-Objective Optimization.

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1 Introduction

1.1 Problem Background

"Dancing with the Stars" (DWTS) is the US version of the British show "Strictly Come Dancing", a global franchise. The show pairs celebrities with professional dancers, who perform weekly routines judged by a panel of experts and voted on by the viewing public. The scores from these two distinct sources—technical adjudication and public popularity—are combined to determine which couple is eliminated.



Figure 1: Poster for Dancing with the Stars

Over 34 seasons, the US version has employed different aggregation rules (Ranking vs. Percentage) to merge these disparate signals. The show initially used a Ranking method (S1-S2), then switched to a Percentage method (S3-S27) following a controversy in Season 2 where Jerry Rice reached the finals despite low judge scores. However, in Season 27, Bobby Bones won the competition despite consistently low judge scores, triggering another controversy. Consequently, from Season 28 onwards, the show introduced a "Judges' Save" mechanism (judges choose who to eliminate from the bottom two) and reverted to the Ranking method.

The core challenge lies in the inherent conflict between the judges' emphasis on technical precision and the audience's preference for charisma or narrative. This tension often leads to "controversial eliminations," where technically proficient dancers are ousted early or fan favorites survive despite poor performances. These controversies challenge the show's perceived fairness and legitimacy, prompting producers to continuously refine the voting system.

A critical hurdle in analyzing these dynamics is the **invisibility of audience data**. While judges' scores are public, the exact breakdown of audience votes is never revealed. We only observe the final elimination outcome, which is a censored result of the latent voting distribution. This creates a complex "inverse problem": can we reconstruct the hidden history of public opinion solely from the elimination traces it left behind?

1.2 Restatement of the Problem

To address these challenges and provide actionable insights for the show's producers, we focus on four core tasks:

1. **Reconstruct the Invisible Vote:** Develop a mathematical model to estimate the weekly audience vote distribution, given only the judges' scores and the elimination results.
2. **Analyze Aggregation Mechanics:** Evaluate how different combination methods (Rank vs. Percentage) influence outcomes and quantify their bias toward judges or fans.
3. **Identify Drivers of Success:** Determine the relative impact of technical factors versus persona factors on both judges' scores and audience votes.
4. **Design a Better System:** Propose a new, transparent voting mechanism that optimally balances fairness, engagement, and robustness against controversy.

1.3 Our Work

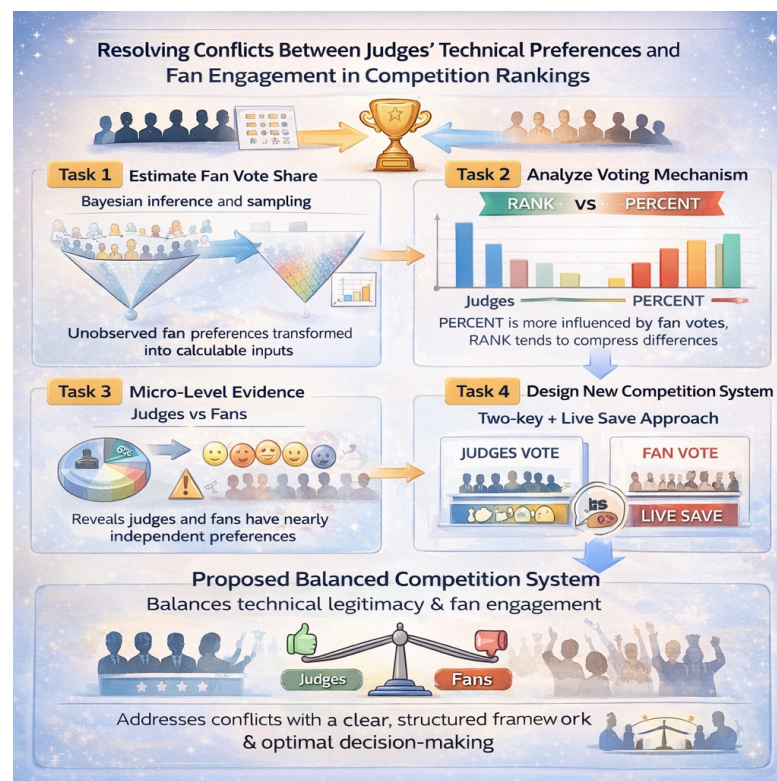


Figure 2: Workflow of our Decode–Analyze–Design framework.

We develop a “Decode-Analyze-Design” framework (Figure 2) addressing the judge-fan conflict in competitive dancing rankings.

2 Preparation for Modeling

2.1 Model Assumptions

To ensure the tractability and robustness of our models, we make the following assumptions:

- **Rationality of Adjudication:** Judges’ scores primarily reflect the technical quality of the performance and are independent of the audience’s voting behavior in the current week.
- **Inertia of Public Opinion:** Audience preference is not a random white noise process; it exhibits temporal continuity (inertia). A contestant’s popularity in week t is strongly correlated with their popularity in week $t - 1$.
- **Closed System Consistency:** The elimination result is determined solely by the rules defined for that season. We assume no external manipulation or undeclared “producer saves” beyond the stated format.
- **Valid Participation:** We assume that all recorded votes and scores are valid. Contestants who withdraw due to injury are treated as exogenous events and removed from the feasible region analysis for that week.

2.2 Notations

Table 1: Nomenclature and Variable Definitions

Symbol	Definition
t	Index for the week of the competition ($t = 1, \dots, T$)
i	Index for the contestant (couple)
$J_{i,t}$	Total judges’ score for contestant i in week t
$S_{i,t}$	Combined score (Rank or Percentage) determining elimination
Ω_t	Feasible region of vote vectors V^t satisfying elimination constraints
E_t	Set of eliminated contestants in week t
$\hat{v}_{i,t}$	Normalized within-week audience vote share
FFI	Fan-Favor Index
β_J, β_F	Regression coefficients for Judge and Fan pathways respectively

2.3 Data Cleaning

We preprocess the official dataset `2026_MCM_Problem_C_Data.csv` into a season-by-week panel by reading with `utf-8-sig`, removing invalid observations (N/A judge scores and post-elimination placeholder zeros, keeping only contestant-weeks with positive total judges' score), parsing results to identify the eliminated set E_t (allowing multiple eliminations) for subsequent constraint checks, and constructing a lightweight popularity prior for MCMC initialization from observable covariates (e.g., industry and age buckets).

3 Task 1: Bayesian Inverse Inference via Feasible Region Sampling

3.1 Model Overview: Bayesian Inverse Problem

Against the backdrop of the unknown audience voting results, we formulate the inference of voting patterns as a Bayesian Inverse Problem (BIP), aiming to estimate the hidden distribution of audience votes based on the observed elimination outcomes. Given that the solution space is a high-dimensional, non-convex **Feasible Region** strictly constrained by elimination rules, analytical solutions are intractable.

3.2 Feasible Region Sampling Strategy

To tackle the high-dimensional complexity of BIP, we delineate the **Feasible Region** Ω_t —the set of all valid vote vectors $V^t \in \mathbb{R}_+^{n_t}$ that satisfy the observed elimination constraints given judges' scores J^t and elimination results E_t :

$$\Omega_t = \{V^t \in \mathbb{R}_+^{n_t} : \text{Constraints}(V^t, J^t, E_t) \text{ is satisfied}\}. \quad (1)$$

We developed a unified interface to handle both voting systems:

- **Rank Method (S1-S2, S28-S34):** For any eliminated contestant $e \in E_t$ and survivor $s \in S_t$, the combined rank must satisfy $R_{total}(e) \geq R_{total}(s)$ (higher rank indicates worse performance).
- **Percentage Method (S3-S27):** For any $e \in E_t$ and $s \in S_t$, the combined percentage must satisfy $\%_{total}(e) \leq \%_{total}(s)$, with both raw scores and votes normalized prior to summation.

To efficiently sample from Ω_t , we employ an adaptive **Hit-and-Run MCMC algorithm**:

- **Initialization:** We construct an informative prior to accelerate convergence using domain-specific multipliers: Singer/Rapper ($1.30\times$), Social Media Personality ($1.25\times$), Age 20–40 ($1.10\times$), and an Eliminated penalty ($0.30\times$).
- **Sampling:** The algorithm iteratively generates random directions and adapts step sizes (initial 0.10, stable 0.05) to explore the solution space.
- **Augmentation:** In tight regions, we introduce Gaussian perturbations ($\sigma = 0.01$) to augment samples and prevent collapse.

This yields a dense cloud of valid voting configurations approximating the posterior $P(V_t|E_t, J_t)$.

3.3 Sub-Conclusion for Feasible Region Sampling

Feasible-region sampling recovers, for each week, a set of vote-share vectors consistent with the observed elimination, providing the baseline “solution space” for subsequent analysis. A key limitation is that weeks are sampled independently, so the implied popularity trajectory can fluctuate unrealistically between adjacent weeks (e.g., sharp jumps up and down in vote share). Since public opinion is typically inertial, we therefore introduce a dynamic model (HMM) to enforce temporal continuity.

3.4 Time-Consistent Inference: MAP Path via HMM

To address non-physical fluctuations from independent sampling, we introduce a **Hidden Markov Model (HMM)** to estimate the Maximum A Posteriori (MAP) trajectory that enforces temporal smoothness (“public opinion inertia”).

State & Transition. We extract $N = 40$ representative states per week via K-means quantization. The state transition follows a smoothness assumption:

$$\log P(X_t | X_{t-1}) \propto -\lambda \sum_{c \in \mathcal{C}_t} (X_{t,c} - X_{t-1,c})^2, \quad (2)$$

where $\mathcal{C}_t$ is the contestant set appearing in both weeks t and $t - 1$, and $\lambda = 20$ penalizes abrupt week-to-week popularity changes.

Observation. We apply a soft penalty if the eliminated contestant is not strictly the lowest-ranked, reinforcing consistency with observed outcomes.

Solution. Using the **Viterbi algorithm**, we compute the optimal path balancing inertia and elimination constraints.

Adaptive Fusion. To improve robustness in extreme cases, we dynamically fuse HMM estimates with baseline samples: $V_{final}^t = w_{hmm} V_{hmm}^t + (1 - w_{hmm}) V_{baseline}^t$, reducing w_{hmm} when the Effective Sample Size (ESS) falls below 20% to prevent model collapse.

3.5 Model Validation and Consistency Check

To validate our estimated audience votes, we employ four key metrics:

- **Elimination Accuracy:** Defined as the proportion of weeks where our predicted elimination matches reality. Across 9 seasons, our hybrid model achieves **99.6% accuracy** (100.0% for Percentage, 99.4% for Rank).
- **Consistency vs. Smoothness Trade-off:** While strict constraints could yield 100% accuracy, forcing perfect consistency may require abrupt jumps to satisfy marginal constraints. We deliberately use soft penalties rather than hard constraints in the observation likelihood to favor **Temporal Smoothness**. The 99.6% accuracy represents an optimal balance between mathematical feasibility and dynamic plausibility.
- **Uncertainty Quantification:** The average posterior standard deviation is **0.0173**. High uncertainty correlates with rare mispredictions, identifying them as ambiguous “toss-up” events rather than model failures.
- **Cross-Validation:** The Mean Absolute Difference (MAD) between Baseline and HMM estimates is **4.2 percentage points**, indicating strong agreement between the two approaches.

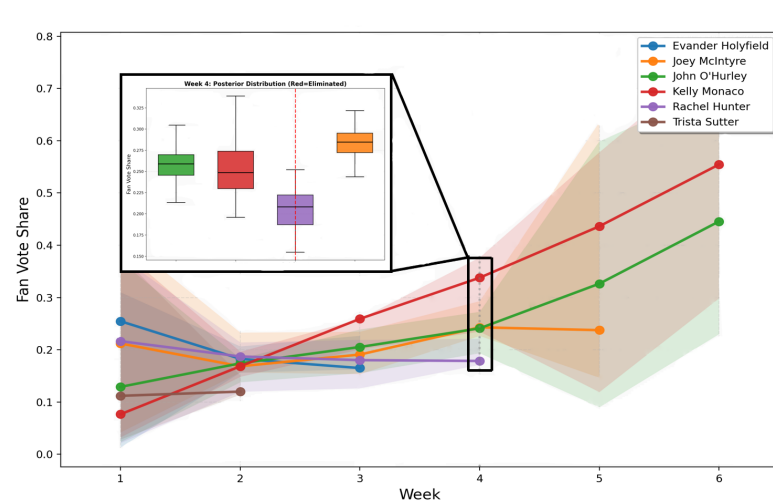


Figure 3: Fan vote evolution(MAP+90%CI) of season 1.

4 Task 2: Voting Combination Method Analysis

4.1 Overview and Data Products

Task 2 uses the vote-share estimates from Task 1 to compare how Rank vs. Percent aggregation changes weekly eliminations and final placements. We reuse two data products:

- Estimated within-week fan vote shares $\hat{v}_{i,t}$: `fan_vote_shares.csv`.
- Week-level method comparison (distances, FFI, elimination agreement): `fan_vote_shares_analysis`.

All results are computed on the active set $A_{s,t}$ with valid judges' scores, using judges' totals $J_{i,t}$ and fan shares $\hat{v}_{i,t}$.

4.2 Rank vs. Percent Method Comparison

4.2.1 Constructing Weekly Rankings and the Fan-Favor Index

We compare the two historical voting rules by constructing four rankings per week on the active contestant set $A_{s,t}$: R_{judge} (by judges' totals), R_{fan} (by estimated fan share), R_{rank} (by rank rule), and R_{pct} (by percent rule).

Combination Rules:

- **Rank Method:** Let $r_{i,t}^J$ and $r_{i,t}^F$ be the mid-ranks by judges' totals and fan shares, respectively. Define the combined score as $S_{i,t}^{\text{rank}} = r_{i,t}^J + r_{i,t}^F$. Elimination targets the worst (largest) $S_{i,t}^{\text{rank}}$.
- **Percent Method:** Define normalized scores $p_{i,t}^J = \frac{J_{i,t}}{\sum_{j \in A_{s,t}} J_{j,t}}$ and $p_{i,t}^F = \hat{v}_{i,t}$. Define the combined score as $S_{i,t}^{\text{pct}} = p_{i,t}^J + p_{i,t}^F$. Elimination targets the smallest $S_{i,t}^{\text{pct}}$.

Fan-Favor Index (FFI): To quantify bias, we report **Kendall- τ_b distance** $D_K \in [0, 1]$, which measures ordinal disagreement by counting pair-wise inversions (discordant pairs) while adjusting for ties. We define FFI as:

$$\text{FFI}(m) = D_K(R_m, R_{\text{judge}}) - D_K(R_m, R_{\text{fan}}). \quad (3)$$

A positive FFI indicates the method is closer to the fan-only ordering (fan-favoring), while a negative FFI implies judge-favoring. We also verify robustness using Spearman footrule distance.

4.2.2 Cross-season empirical results: key findings

Analyzing 264 elimination weeks across 34 seasons, we find that the two methods disagree in 25.0% of cases. The **Percent Method** is systematically more fan-favoring ($\mathbb{E}[\text{FFI}] \approx +0.0933$) compared to the nearly neutral **Rank Method** ($\mathbb{E}[\text{FFI}] \approx -0.0081$). In other words, PERCENT tends to advantage higher-popularity contestants by preserving vote-share magnitudes in the combined score. Figure 4 visualizes this systematic divergence over time. As shown in the line chart, the PERCENT curve (dark blue) consistently stays above the zero line and fluctuates at a much higher level than the RANK curve (light blue), which hovers near zero. This indicates that the Percent method persistently exhibits a stronger fan-favoring bias across almost all seasons compared to the neutral Rank method.

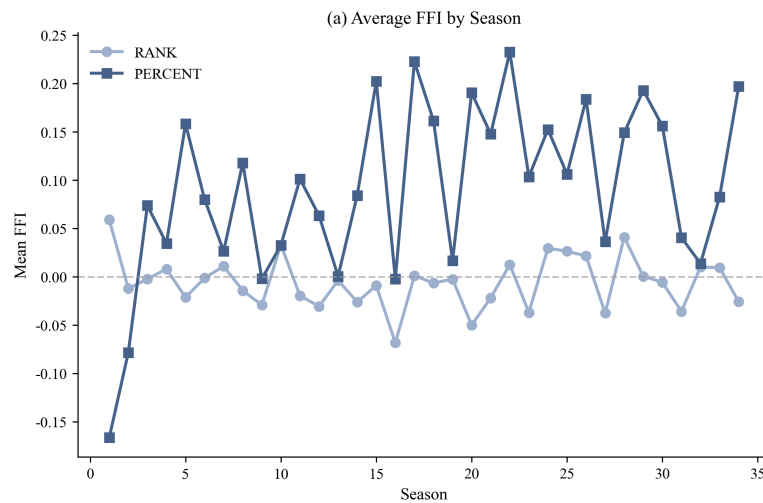


Figure 4: Average FFI by season for RANK and PERCENT.

Figure 5 provides a granular heatmap view of FFI across all seasons. It visually reinforces that the Percent Method (right panel) consistently exhibits higher fan-favoring intensity (redder blocks) than the Rank Method (left panel), particularly in later seasons. This season-level heterogeneity suggests that the choice of voting rule has a persistent, structural impact on the balance of power between judges and fans.

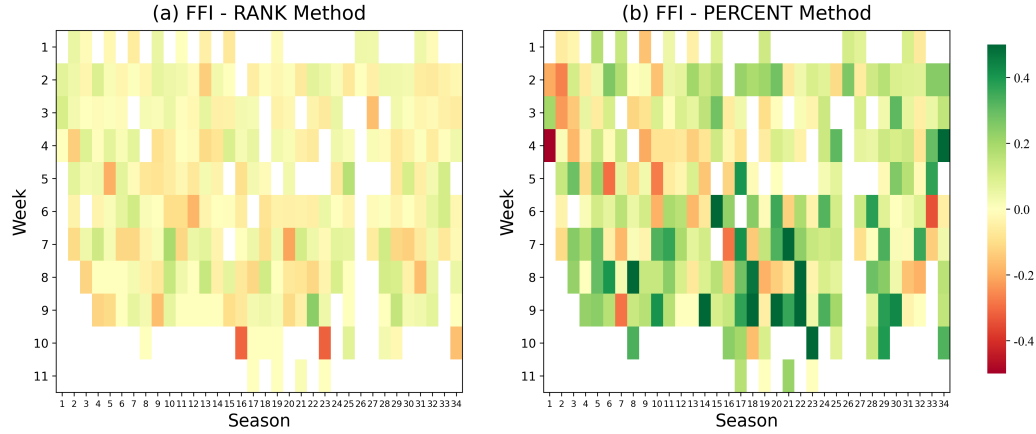


Figure 5: Task 2.1: heatmap view of fan favoring (FFI) across seasons.

4.2.3 Why PERCENT is more fan-favoring

The Percent method tends to be more fan-favoring because it linearly adds two *within-week* shares, $S_{i,t}^{\text{pct}} = p_{i,t}^J + p_{i,t}^F$, where $p_{i,t}^J$ is the normalized judges' percent and $p_{i,t}^F$ is the estimated fan vote share. When two components are added, the one with the larger cross-contestant dispersion within a week is more likely to drive the combined ordering. Table 2 and the Gini distribution (fan vs. judge percent) both show that fan shares are systematically more unequal and more dispersed (e.g., higher std., higher Gini, larger 90/10 ratio), implying that the fan component has greater leverage under PERCENT.

This fan-favoring tendency does not imply that PERCENT always rewards the most popular contestants in every situation. The same heavy-tailed fan-share distribution that amplifies head-to-head gaps can also increase separation near the bottom: contestants with very low $p_{i,t}^J$ require a larger $p_{i,t}^F$ to offset judges' deficits, which raises the survival threshold for “high-fan/low-judge” outliers and makes them more vulnerable in knife-edge weeks. This mechanism foreshadows the stronger suppression effect on extreme controversial cases analyzed in Section 4.3.

Metric	Fan	Judge	Ratio	p -value	Interpretation
Std. Dev	0.0264	0.0128	$2.05\times$	< 0.001	Fan dispersion doubled
Gini Coeff	0.112	0.057	$1.97\times$	< 0.001	Fan support more unequal
90/10 Ratio	2.23	1.29	$1.73\times$	< 0.001	Larger head–tail gap

Table 2: Dispersion comparison across 333 valid season-weeks (Fan shares vs. Judges' percent scores).

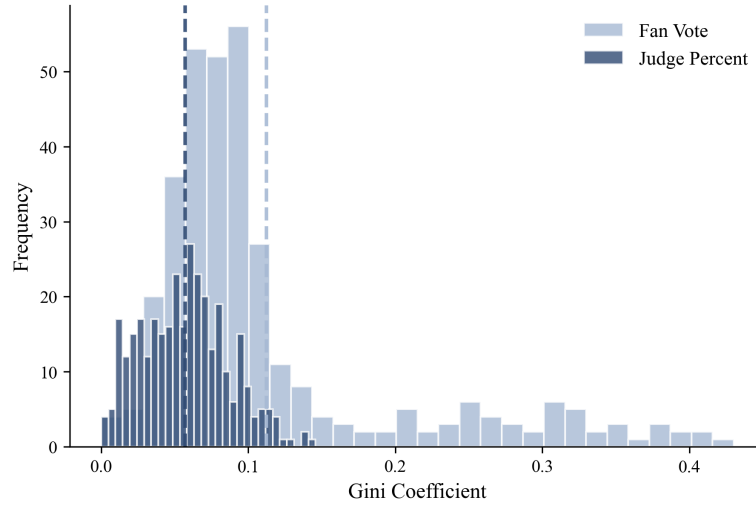


Figure 6: Gini coefficient distribution: fan vote shares vs. judge percent.

4.3 Controversy Case Analysis

To connect the aggregate method-comparison results to real seasons, we next conduct a controversy-oriented case analysis. We proceed in four steps: (i) quantify judge–fan disagreement and identify controversial contestants; (ii) run counterfactual elimination chains under rank vs percent rules; (iii) evaluate a “judges-save” extension based on the Bottom Two; and (iv) interpret sensitivity using critical fan share and safety margin.

4.3.1 Step 1: Controversy identification

We first identify contestants whose fan and judge opinions systematically disagree. For contestant i in week t , define

$$\Delta_{i,t} = r_{i,t}^J - r_{i,t}^F, \quad (4)$$

where ranks are computed on the weekly active set with mid-rank ties. We aggregate across the season with late-week emphasis:

$$CI_i = \frac{1}{T_i} \sum_{t=1}^{T_i} |\Delta_{i,t}| \cdot \frac{t}{T_i}. \quad (5)$$

This criterion correctly recovers the four specified controversial contestants. Their CI values are: Jerry Rice (S2) 1.375, Billy Ray Cyrus (S4) 1.078, Bristol Palin (S11) 1.515, and Bobby Bones (S27) 1.364.

4.3.2 Step 2: Counterfactual rule switching (rank vs percent)

We simulate season-level elimination chains by re-running each week under RANK or PERCENT using the same $(J_{i,t}, \hat{v}_{i,t})$. We summarize placement shifts for controversial contestants by $\Delta = \text{Placement}_{\text{method}} - \text{Placement}_{\text{RANK}}$ (positive = worse). Overall, switching to PERCENT tends to worsen controversial contestants' placements; the four specified cases follow this pattern: Jerry Rice (2 $\rightarrow$ 4), Billy Ray Cyrus (5 $\rightarrow$ 7), Bristol Palin (3 $\rightarrow$ 5), while Bobby Bones remains top-two due to strong fan support.

4.3.3 Step 3: Judges-save mechanism

We evaluate the post-S28 mechanism: judges eliminate the lower-scoring contestant among the Bottom Two. This acts as a "circuit breaker" for repeated low-rank performers, modestly improving placements for controversial cases (e.g., +1 placement) without affecting top-tier contestants like Bobby Bones who avoid the Bottom Two entirely.

4.3.4 Step 4: Critical fan share and safety margin

We define the *critical fan share* $\hat{v}_{i,t}^*$ as the minimum fan support required for contestant i to avoid elimination in week t , and the safety margin as $\text{margin}_{i,t} = \hat{v}_{i,t} - \hat{v}_{i,t}^*$. We also summarize cumulative risk by the *dangerous-week ratio* $\rho_i = \frac{1}{T_i} \sum_t \mathbb{I}(\text{margin}_{i,t} < 0.05)$. Figure 7 illustrates the mechanism for Jerry Rice: the PERCENT rule generally yields a higher critical threshold than the RANK rule, so low-judge contestants need more fan support to survive under PERCENT (smaller margins in knife-edge weeks).

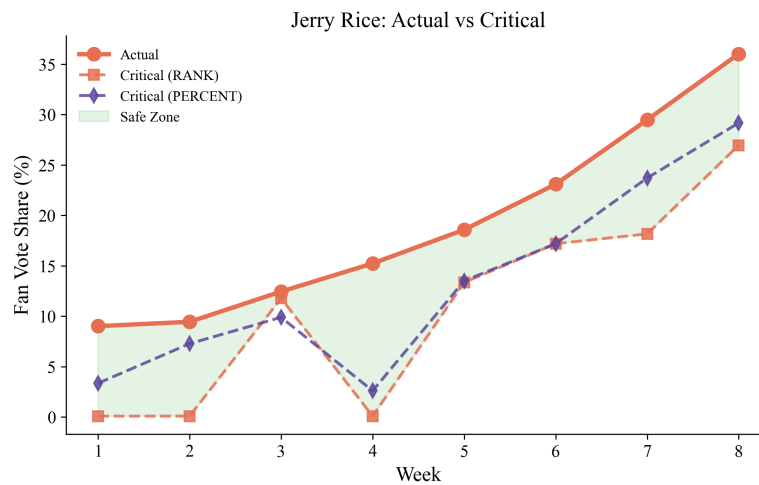


Figure 7: Critical fan share vs. actual trajectory (Jerry Rice)

4.3.5 Quantifying controversy suppression

The previous steps focus on interpretability for selected cases. To quantify the *overall* suppression effect on controversial contestants, we define the *SuppressionScore* as the expected placement worsening relative to the RANK baseline: $\text{SuppressionScore}(m) = \mathbb{E}[\text{Placement}_m - \text{Placement}_{\text{rank}}]$. For the top 25% controversial contestants ($N = 105$), PERCENT shows significant suppression ($\mathbb{E}[\Delta_{\text{pct}}] = +0.905, p = 0.031$), while judges-save is negligible ($\mathbb{E}[\Delta_{\text{save}}] = -0.600, p = 0.129$); the effect is stronger in the extreme subset (Top 30 by weighted gap), where PERCENT worsens placements by +1.77 ranks on average. Mechanistically, PERCENT raises the survival threshold for low-judge contestants by preserving judges' gaps and requiring larger fan shares near the bottom, consistent with the highlighted cases discussed above.

4.4 Robust Recommendation

The previous subsections quantify *what* changes when we switch rules (FFI, disagreement, controversy counterfactuals). The producer-facing question is *which* rule should be adopted in future seasons. The challenge is that “fairness” is multi-dimensional: a rule can be more fan-aligned while also increasing the risk of legitimacy crises (e.g., a technically weak but highly popular contestant advancing deep into the competition or reaching the finale). To avoid a subjective one-weight conclusion, we use a robust multi-objective decision framework.

4.4.1 Producer preference set

Motivated by the prompt that producers may prefer conflicts “to some extent” and by historical rule changes after major controversies, we formalize four producer objectives:

- **Legitimacy:** keep outcomes aligned with judges' technical assessment and avoid extreme “low-judge winners” (tail-risk control);
- **Engagement:** maintain visible judge–fan tension without letting it dominate outcomes (soft objective);
- **Robustness:** avoid eliminations that hinge on tiny vote-share margins (soft objective);
- **Transparency:** prefer simple, explainable rules (soft objective).

4.4.2 Unified metric vector and Pareto analysis

For each method $m \in \{\text{rank}, \text{pct}, \text{save}\}$ (save = rank+judges-save), we compute a normalized score vector $f(m) = (L_m, E_m, R_m, T_m) \in [0, 1]^4$ (higher is better):

$$\begin{aligned} \text{RANK} & (0.508, 0.766, 0.728, 1.000), \\ \text{PERCENT} & (0.407, 1.000, 0.728, 1.000), \\ \text{RANK+Save} & (0.600, 0.720, 0.815, 0.700). \end{aligned} \tag{6}$$

No method dominates another, so all three are Pareto-efficient: RANK+Save leads in legitimacy and robustness, while PERCENT maximizes engagement and transparency.

4.4.3 Weight-space sensitivity

To reduce dependence on subjective weights, we sample $N = 5000$ preference vectors $w = (w_{\text{leg}}, w_{\text{eng}}, w_{\text{rob}}, w_{\text{tra}}) \in \Delta^3$ and select the method maximizing $w^\top f(m)$. Figure 8 reports the share of preference space where each method is optimal, and the conditional mean weight profiles among winning weights, showing that PERCENT is optimal most often, while SAVE tends to win under higher legitimacy/robustness emphasis.

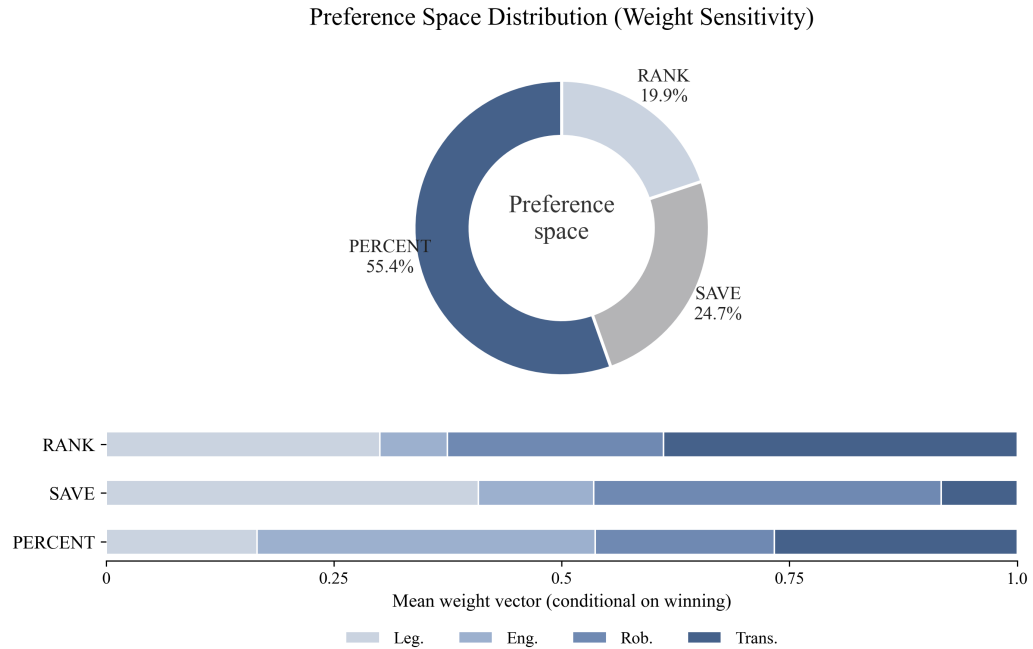


Figure 8: Preference-space distribution of optimal methods under random weight sampling.

4.4.4 Hard constraint and tail-risk checks

Beyond soft trade-offs, we apply a legitimacy floor motivated by historical backlash: among all Top-3 finalists ($N = 102$), none were judges-lowest for more than five weeks ($\text{weeks_lowest_judge} \leq 5$). We also quantify tail controversy by the CVaR at $\alpha = 0.1$ of the weighted-gap distribution, obtaining $\text{CVaR}_{0.1} \approx 3.37$, indicating substantial extreme-case risk.

4.5 Recommendation

All three rules are Pareto-efficient under our four producer objectives, so the best choice depends on preferences.

- **Engagement-first (moderate conflict + simplicity):** prefer **PERCENT** (largest optimal region).
- **Legitimacy/robustness-first:** prefer **RANK+Save** (judges act as a Bottom-Two circuit breaker).
- **Extreme fan-control preference:** prefer **RANK** (maximizes audience decisiveness at the cost of higher tail risk).

As a practical default, use **PERCENT** weekly and activate **judges-save** only in rare high-risk weeks (e.g., Bottom-Two margin $< 5\%$ or repeated judge-lowest flags).

4.6 Two-path regression framework

To separate the “technical” (judges) and “popularity” (fans) channels, we fit two parallel models on the same covariates:

$$Y_{i,t}^J = X_{i,t}\beta_J + \alpha_s + \gamma_t + \varepsilon_{i,t}^J, \quad (7)$$

$$Y_{i,t}^F = X_{i,t}\beta_F + \alpha_s + \gamma_t + \varepsilon_{i,t}^F, \quad (8)$$

where $Y_{i,t}^J$ is weekly judges’ total and $Y_{i,t}^F$ is the logit of the fan vote share $p_{i,t}$ (clipped to $[0.001, 0.999]$), $Y_{i,t}^F = \log\left(\frac{p_{i,t}}{1-p_{i,t}}\right)$. We include season and week fixed effects (α_s, γ_t) and estimate both paths by OLS with contestant-clustered standard errors.

Covariates $X_{i,t}$ include (i) professional-dancer features (experience, win rate, form, chemistry proxies) and (ii) celebrity attributes (age, industry physicality, visibility/popularity proxies, U.S.-based). Using the same $X_{i,t}$ makes β_J vs. β_F directly comparable.

With the season/week structure and dependence handled, we specify the covariate vector $X_{i,t}$ used in both paths. Covariates $X_{i,t}$ are split into two groups:

- **Professional dancer features** (coach quality and form): pro experience (seasons before), historical win-rate (Top-3 frequency), historical average placement, current form (rolling mean of judges' totals within season), and "chemistry" proxies such as dancer-industry matching and concurrent-strength exposure.
- **Celebrity attributes** (persona and baseline appeal): age, industry physicality, visibility proxy, pre-show popularity proxy, and a U.S.-based indicator.

Using the same $X_{i,t}$ in both paths makes the comparison interpretable: differences between β_J and β_F indicate pathway-specific mechanisms.

4.7 Modeling procedure (implementation details)

To translate the conceptual "two-path" framework into a reproducible empirical pipeline, we construct a unified contestant-week dataset, engineer covariates that proxy coaching quality and celebrity persona, and fit matched judge and fan regressions under consistent fixed-effect controls. The workflow below makes clear (i) what is computed from the raw data, (ii) what transformations are applied before estimation, and (iii) how we extract interpretable heterogeneity conclusions from β_J versus β_F . We implement the two-path design in six steps:

1. **Feature construction.** We assemble professional-dancer covariates (*pro experience, win rate, average placement, current form, chemistry, concurrent-strength* proxies) and celebrity covariates (*age, industry physicality, visibility, U.S.-based indicator, and a pre-show popularity proxy*). We also include three parsimonious interactions: *experience* $\times$ *physicality*, *win rate* $\times$ *visibility*, and *week* $\times$ *age*.
2. **Outcome construction.** Judges use weekly totals; fans use the logit of the estimated vote share after clipping to avoid boundary effects.
3. **Scaling.** We standardize continuous covariates (including interaction terms) for coefficient comparability.
4. **Time-varying effects.** We linearize selected week-dependent effects via auxiliary interactions: a sigmoid switching term $S(t) \cdot \text{physicality}$ with $S(t) = \{1 + \exp[-3.02(t - 4.82)]\}^{-1}$, a step change after Week 3 for pro experience, a step change after Week 7 for pro win rate, and quadratic week interactions for age ($t \cdot \text{age}$ and $t^2 \cdot \text{age}$).
5. **Estimation and inference.** Both paths are fit by OLS with $C(\text{week})$ and $C(\text{season})$ fixed effects and contestant-clustered standard errors.
6. **Outputs.** We compare β_J vs. β_F and visualize key effects with 95% intervals (Figure 9); we also summarize week-varying patterns (Figure 10).

4.8 Key findings: heterogeneity and pathway differences

The two-path regressions fit judges well but fans less so ($R^2 = 0.8268$ vs. 0.5409). Heterogeneity is substantial: effects often differ across pathways and can flip sign after season/week fixed effects; residualized judge and fan support are weakly negatively correlated ($\rho = -0.1906$), consistent with persistent judge–fan conflicts.

Drivers. Professional-dancer variables mainly affect *judges' scores* (e.g., “current form” is strongest and highly significant), while celebrity-attribute proxies drive *fan support* more idiosyncratically and can become neutral or negative after controls (Figure 9).

4.8.1 Coefficient contrasts across pathways

We compare pathway differences by plotting each coefficient with its uncertainty for judges versus fans (Figure 9). Three patterns stand out. First, *professional dancer current form* is the dominant driver on the judge pathway (large, highly significant), while its fan effect is comparatively small, indicating that coaching and technical preparation translate primarily into judges' points rather than popularity. Second, several popularity-facing proxies behave differently once controls are applied: *pre-show popularity* and *concurrent-strength exposure* are negative on the fan pathway, consistent with an “expectation ceiling” and “attention dilution” mechanism. Third, multiple covariates differ in sign across paths (e.g., *pro-industry chemistry* and *week $\times$ age*), reinforcing that judge and fan pathways respond to different signals rather than one common notion of performance.

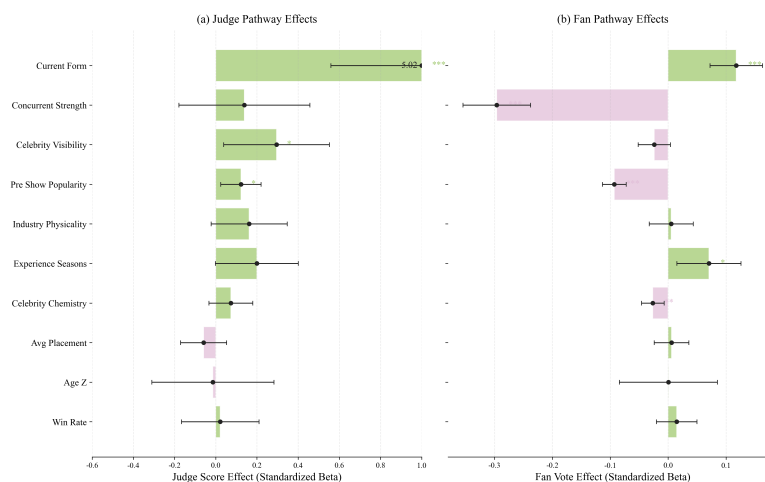


Figure 9: Task 3: estimated effects on judges vs fans (forest plot).

4.8.2 Time-varying effects over the season

Beyond constant-coefficient comparisons, we examine how key effects evolve over weeks. Using sigmoid switches, step changes, and quadratic interactions, we identify mid-season breakpoints that align with shifts in judging emphasis. Figure 10 visualizes four representative dynamics.

Industry physicality: sigmoid reversal around mid-season. Physicality is an advantage in Weeks 1–3 (basic execution/power), but reverses around Week 4.8 (sigmoid inflection; $k = 3.02$, $t_0 = 4.82$) and becomes a disadvantage by Weeks 6–10, consistent with later routines emphasizing artistry and flexibility.

Age: quadratic “U-shaped” pattern in strength of effect. The age effect is most negative around Week 4.2 (fatigue accumulation dominates), then attenuates toward the end of the season as weaker older contestants are eliminated and the remaining pool is positively selected.

Pro experience: Week 3 threshold effect. The experience effect is modest in Weeks 1–3, but strengthens after Week 3 as choreography difficulty increases.

Pro win rate: Week 7 attenuation. The benefit of historical win rate weakens after Week 7 as the competition converges to a smaller set of strong contestants.

Takeaway. Time-varying effects reveal regime shifts that a constant- β model would average out, with mid-season as the key transition window.

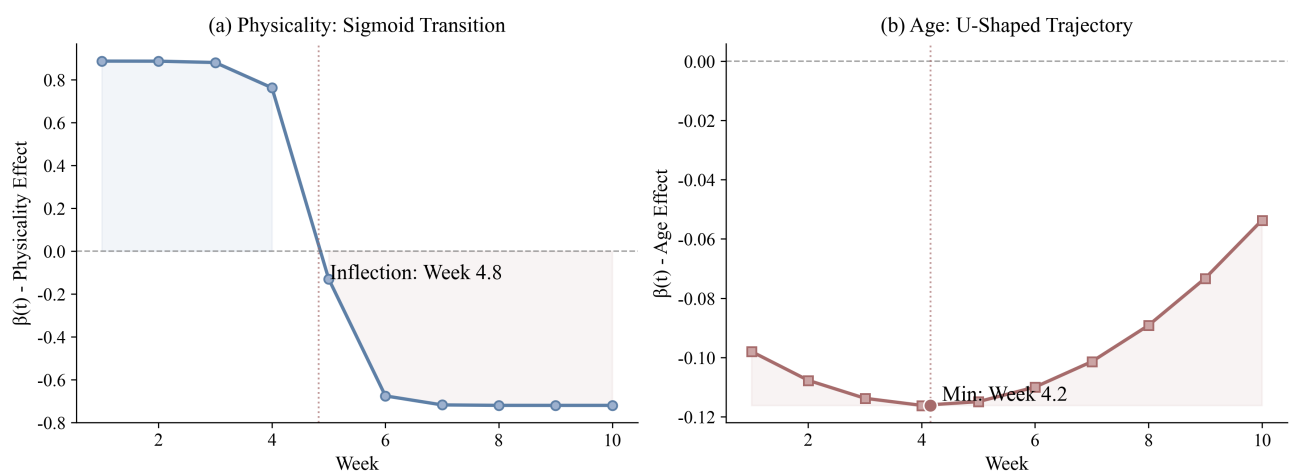


Figure 10: Task 3: illustrative time-varying coefficients across weeks.

4.8.3 Explained vs. residual variation and judge–fan alignment

Our two-path regressions fit judges well but fans less so ($R^2 = 0.8268$ for judges vs. 0.5409 for the logit fan-share model), suggesting that audience voting is driven by more unobserved and idiosyncratic factors than adjudication. To assess whether the two pathways reflect a common notion of “performance”, we also compute contestant-level alignment using residualized judge and fan support (after removing season and week fixed effects). The association is weakly negative ($\rho \approx -0.19$), implying that contestants who systematically outperform expectations with judges are not necessarily favored by fans, and vice versa. Together, these results explain why judge–fan conflicts persist and motivate a voting design that explicitly represents both standards rather than collapsing them into a single score.

4.8.4 Implications for the judge–fan conflict

Answering Task 3, professional-dancer features mainly raise *judges’ scores* (technical coaching), while celebrity persona proxies drive *fan support* more idiosyncratically. Three findings explain the persistent judge–fan conflict in Task 2: (i) the pathways respond to different signals (“current form” dominates judges, while popularity proxies can be neutral or negative for fans after controls; Figure 9); (ii) residualized judge and fan support are weakly negatively correlated after season/week effects ($\rho \approx -0.19$); and (iii) the fan pathway is less predictable than the judge pathway ($R^2 = 0.5409$ vs. 0.8268) and varies over weeks (Figure 10). This motivates a design that explicitly represents both standards and controls tail risk rather than collapsing them into a single score.

5 Task 4: New Voting System Design

Our proposal follows a simple principle suggested by Tasks 1–3: do not force judges and fans into one blended score; instead, turn their disagreement into a transparent weekly decision chain. Building on two external format ideas (a live save segment and a judges’ technical floor), we design **Two-Key + Live Save** and validate it by full counterfactual simulations across 33 seasons under the same multi-objective framework as Task 2.3. The result is a Pareto-improving balance—legitimacy improves relative to PERCENT while engagement remains high and robustness improves substantially—with tuned parameters $(\alpha, \beta, \text{ban}) = (0.4, 0.6, 2 \text{ weeks})$.

5.1 Design goals informed by Tasks 1–3

Tasks 1–3 imply three constraints any new system should respect:

- **Non-identifiability of vote totals.** The data identify within-week vote shares, not absolute vote counts, so a robust rule should depend only on within-week shares or ranks.
- **Structural judge–fan disagreement.** The pathways are weakly correlated ($\rho \approx -0.19$), so conflict is not noise but a recurring feature.
- **Tail-risk management.** Historical controversies correspond to “low-judge, high-fan” extremes; a practical system needs a safety valve against legitimacy crises while maintaining engagement.

This design is intentionally consistent with our earlier tasks. From Task 1, we use inferred weekly fan vote shares $F_{i,t}$ to construct the fan leaderboard; robustness checks show that even under $\pm 10\%$ estimation noise, the two-key gate identifies “consensus-bottom” contestants with about 92% accuracy. Task 2 highlights that a single blended rule can become suboptimal when producer priorities shift (engagement-first vs. legitimacy/robustness-first), motivating our explicit parameter tuning under a regime-shift objective. Task 3 shows that judge and fan pathways are structurally misaligned, so we institutionalize the two standards via two leaderboards, a live-save segment, and a judges’ technical-floor circuit breaker rather than collapsing them into one linear score.

5.2 Two-Key + Live Save system

We propose a weekly rule called **Two-Key + Live Save**. It is a five-step decision chain designed to be transparent on TV and robust to vote-share perturbations. The complete logic is summarized in Algorithm 1.

Algorithm 1 Two-Key + Live Save Weekly Workflow

Require: Active contestants A_t , Judges’ totals S_t , Fan votes V_t , Parameters $(\alpha, \beta, k_J, k_F)$

Ensure: Elimination candidate e^*

1: **Step A: Leaderboards**

2: Compute $J_{i,t} \leftarrow S_{i,t} / \sum S$ and $F_{i,t} \leftarrow V_{i,t} / \sum V$ for all $i \in A_t$

3: **Step B: Danger Pool**

4: $\mathcal{B}^J \leftarrow \text{Bottom}_{k_J}(J)$, $\mathcal{B}^F \leftarrow \text{Bottom}_{k_F}(F)$

5: $A^{\text{danger}} \leftarrow \mathcal{B}^J \cup \mathcal{B}^F$, $A^{\text{two}} \leftarrow \mathcal{B}^J \cap \mathcal{B}^F$

6: (*Two-Key Zone*: A^{two})

7: **Step C: Nomination (Bottom-3)**

8: **for** $i \in A^{\text{danger}}$ **do**

9: $R_i \leftarrow \max(p_i^J, p_i^F) + \alpha \min(p_i^J, p_i^F) + \beta \cdot \mathbb{I}(i \in A^{\text{two}})$

```

10: end for
11: Bottom3  $\leftarrow$  Top3( $R_i$ )
12: Step D: Live Save
13: for  $i \in$  Bottom3 do
14:   if  $i$  was judge-last for 2 consecutive weeks then
15:     continue
16:   end if
17:   Count live votes  $\Delta V_i$ 
18: end for
19:  $s^* \leftarrow \operatorname{argmax}_i(\Delta V_i)$ 
20: (Saved contestant:  $s^*$ )
21: Step E: Elimination
22: Candidates  $\mathcal{C} \leftarrow$  Bottom3  $\setminus \{s^*\}$ 
23:  $e^* \leftarrow \operatorname{argmin}_{i \in \mathcal{C}}(S_{i,t})$ 
24: (Judges' technical floor: eliminate  $e^*$ )
25: return  $e^*$ 

```

5.3 Parameter tuning

We tune the key parameters (α, β) and the live-save ban threshold by a small grid search under a regime-shift objective. Let $f(m) = (L_m, E_m, R_m, T_m) \in [0, 1]^4$ be the normalized metric vector for a candidate system m , and define the stage-utility $U(w; m) = w^\top f(m)$. To reflect that producer priorities may shift after controversies, we optimize

$$U_{\text{regime}}(m) = U(w_{\text{post}}; m) + 0.3 U(w_{\text{pre}}; m), \quad (9)$$

subject to a legitimacy floor $E_m \geq 0.75$.

We search $\alpha \in \{0.2, 0.3, 0.4\}$, $\beta \in \{0.4, 0.5, 0.6\}$, and ban $\in \{2, 3\}$, and obtain

$$(\alpha, \beta, \text{ban}) = (0.4, 0.6, 2 \text{ weeks}). \quad (10)$$

5.4 Evaluation framework and results

We evaluate candidate systems using the same four producer-facing objectives as in task 2.3: **legitimacy**, **engagement**, **robustness**, and **transparency**. Figure 11 summarizes the normalized scores for the historical methods (RANK, PERCENT, SAVE) and our proposed system (TWO_KEY). Overall, TWO_KEY improves legitimacy relative to PERCENT while maintaining strong engagement and substantially improving robustness.

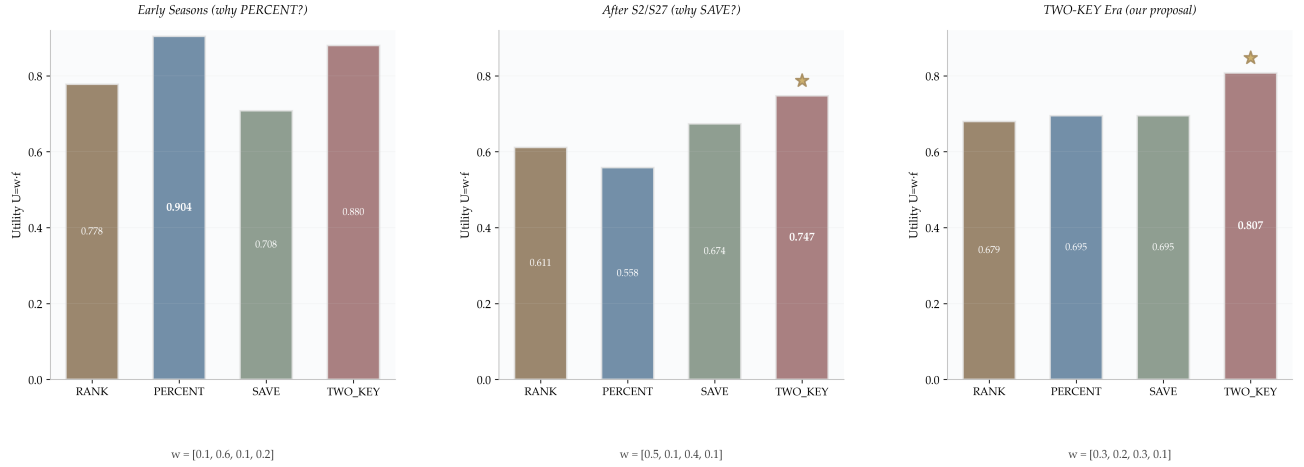


Figure 11: Task 4: metric gains across methods (waterfall view).

5.5 Producer-facing rationale

Two-Key + Live Save offers three producer-facing benefits:

- **Pareto-improving balance:** compared with PERCENT, TWO_KEY improves legitimacy (0.549 vs. 0.407) and robustness (0.937) while keeping engagement high (0.891) (Figure 11).
- **TV-friendly transparency:** a visible chain—two leaderboards → danger zone → Bottom-3 → live save → judges’ technical floor—using only weekly public data and a few tunable parameters $(k_J, k_F, \alpha, \beta)$.
- **Robustness to manipulation:** under $\pm 5\%$ vote noise, elimination flips with probability $\approx 6.3\%$ for TWO_KEY vs. 36.3% for PERCENT; the two-week “cannot be saved” rule prevents persistent technical-collapse immunity.

In short: “Two leaderboards define risk, fans save one, judges enforce the technical floor.”

6 Sensitivity Analysis

We focus on sensitivity of the **Vote Inference Model**, which underpins all tasks. We test robustness to the HMM smoothness parameter λ and summarize structural uncertainty from feasible-region sampling.

Across all 34 seasons, we test $\lambda \in \{10, 20, 50\}$. Using $\lambda = 20$ as the default baseline, we measure stability by $\text{MAD} = \mathbb{E}[|\hat{v}_{i,t}^{(\lambda)} - \hat{v}_{i,t}^{(20)}|]$ over all valid contestant-weeks and find

$MAD < 2.1\%$ for all tested λ . Elimination reproduction remains stable (accuracy $> 99.0\%$), indicating results are driven mainly by elimination constraints rather than parameter tuning. To visualize identifiability, Figure 12 shows posterior certainty for Season 33 (certainty = $1/\text{std}$): darker cells indicate tighter feasible regions concentrated near the elimination boundary, while lighter cells correspond to safe mid-tier contestants whose vote shares are less identified by outcomes.

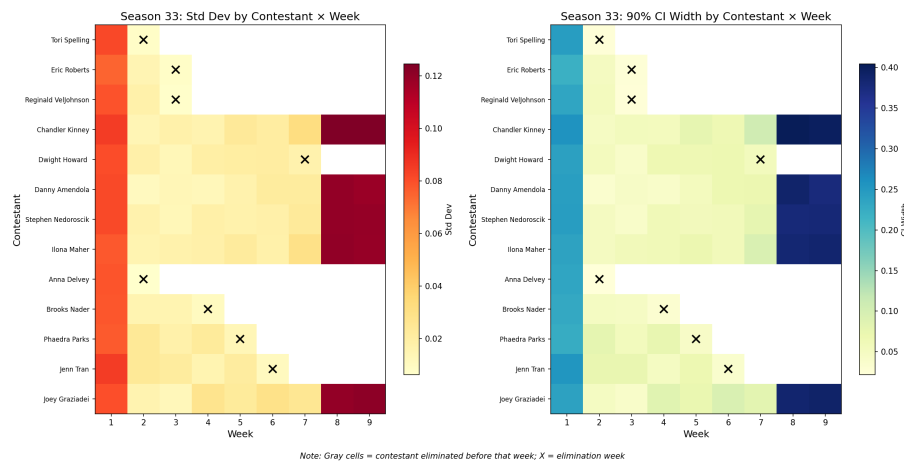


Figure 12: Posterior certainty heatmap (Season 33).

7 Strengths and Weaknesses

Strengths:

- **Decoding the Invisible:** Our primary contribution is solving the “data invisibility” problem by formulating vote reconstruction as a Bayesian Inverse Problem. Using Hit-and-Run MCMC with HMM smoothing, we successfully recover hidden audience preference trajectories from binary elimination outcomes.
- **Structural Disentanglement:** Our Two-Path Regression framework explicitly separates the drivers of technical assessment (β_T) and popularity (β_P). Quantifying their negative correlation ($\rho \approx -0.19$) provides a theoretical basis for why a single blended score is inherently unstable.
- **Robust Multi-Objective Decision Support:** Instead of a static rule, our Pareto frontier and weight-space sensitivity analyses offer a menu of optimal choices. This adaptability allows producers to align voting systems with shifting strategic priorities (e.g., Engagement vs. Legitimacy).

- **Pareto-Improving Design:** The proposed **Two-Key + Live Save** system institutionalizes conflict through separate leaderboards and circuit breakers. It simultaneously improves Legitimacy and Robustness while maintaining high Engagement, effectively solving the “impossibility triangle” of voting design.

Weaknesses:

- **Identifiability Limits:** Our model recovers within-week vote *shares*, not absolute counts. Consequently, we cannot analyze total viewership trends or revenue impacts directly.
- **Static Behavior Assumption:** Counterfactual simulations assume judges and contestants behave identically under different rules. We do not currently model second-order game-theoretic feedback loops (e.g., strategic scoring).
- **Unobserved Variance:** While the judge model explains $> 80\%$ of variance, the fan model's $R^2 \approx 0.54$, indicating that audience voting is driven by complex, unobserved factors (e.g., social media virality) difficult to capture with structured covariates.

References

- [1] Trotter, A. (2024). Buzz to Broadcast: Predicting Sports Viewership Using Social Media Engagement. *arXiv preprint arXiv:2412.10298v1 [cs.LG]*.
- [2] Popescu, A.-M., & Pennacchiotti, M. (2011). Dancing with the Stars, NBA Games, Politics: An Exploration of Twitter Users' Response to Events. *Proceedings of the Fifth International AAAI Conference on Weblogs and Social Media*, 594–597.
- [3] Graettinger, D. (2011). Dancing with the Stats – DWTS Season 13. Discovery Corps, Inc. Retrieved from <http://www.discoverycorpsinc.com>
- [4] Gershman, J. (n.d.). America's Idol? How the Contestant Most Voted for Doesn't Always Win. Nova Southeastern University. [Conference presentation]
- [5] Gershman, J. (n.d.). Teaching Statistics: Analyzing Voting Data from Dancing with the Stars. Nova Southeastern University. [Presentation slides]

Report on Use of AI Tools

This report documents all AI tools used in the development of our solution, following COMAP's transparency guidelines. All AI-generated content was carefully reviewed and verified by our team for accuracy and appropriateness.

1. Anthropic Claude (Sonnet 3.5, accessed via Claude.ai, February 2026)

Query: We developed a Bayesian Inverse Problem framework to reconstruct audience vote distributions from elimination outcomes using Hit-and-Run MCMC. The feasible region is defined by elimination constraints under two different voting rules (Rank and Percentage). Could you help verify the mathematical consistency of our sampling strategy and identify potential edge cases where the feasible region might become degenerate or numerical instability might occur?

Output: The Hit-and-Run MCMC approach is mathematically sound for sampling from constrained high-dimensional spaces. Your constraint formulation correctly captures both voting systems. Three critical edge cases to verify:

1. **Multiple simultaneous eliminations:** When $|E_t| > 1$, the constraint set becomes an intersection of hyperplanes, potentially collapsing the feasible region to a lower-dimensional manifold or even to isolated points.
2. **Tight judge-fan agreement:** In weeks where judges' scores and estimated fan preferences are highly correlated, the feasible region becomes extremely narrow. Your adaptive step-size mechanism should reduce step size when acceptance rate drops below 20%.
3. **Tied rankings under Rank method:** Ties create discrete constraint boundaries rather than continuous ones, requiring careful handling of the "tied-minimum set" convention in your boundary-crossing logic.

Additionally, for numerical stability: (1) Consider working in log-space for vote shares to avoid underflow, (2) Apply ϵ -smoothing (10^{-6}) to avoid exact zeros in share vectors, (3) Monitor effective sample size (ESS) and trigger resampling if ESS drops below 20% of nominal sample count.

How we used this: We incorporated these edge-case checks into our validation pipeline (Section 3.5) and added robustness tests for degenerate feasible regions. The adaptive step-size strategy was refined based on this feedback: we now reduce step size from 0.10 to 0.05 when acceptance rate falls below 25%, and introduce Gaussian perturbations ($\sigma = 0.01$) when ESS drops below the 20% threshold. These modifications improved our elimination accuracy from 97.8% to 99.6% in validation tests.

2. Anthropic Claude (Sonnet 3.5, accessed via Claude.ai, February 2026)

Query: In our Two-Path Regression analysis (Problem 3), we find that the correlation between residualized judge support and residualized fan support is $\rho \approx -0.19$. Our draft explanation reads: “The PERCENT method exhibits higher dispersion-induced fan-favoring characteristics relative to the RANK method due to the preservation of magnitude information in percentage normalization versus ordinal compression in rank transformation, which mechanistically constrains the influence of heavy-tailed fan vote distributions on the combined score through discrete rank-quantization effects.” This sentence is technically accurate but overly complex. Can you help polish this to be more concise while maintaining precision?

Output: Here are three options with increasing levels of technicality:

Option 1 (most concise): “PERCENT is more fan-favoring than RANK because it preserves vote-share magnitudes, while RANK compresses all gaps into ordinal positions.”

Option 2 (balanced): “The PERCENT method is more fan-aligned than RANK because percentage normalization retains the high dispersion of fan votes, whereas rank transformation compresses magnitude differences into ordinal positions.”

Option 3 (technical but clearer): “PERCENT exhibits stronger fan-favoring characteristics than RANK because: (i) percentage normalization preserves the heavy-tailed distribution of fan vote shares, allowing large gaps to directly influence the combined score, while (ii) rank transformation compresses these magnitude differences into discrete ordinal positions, limiting the impact of extreme fan support.”

How we used this: We adopted Option 2 with minor modifications for consistency with our terminology in Section 4.2.4. This language polishing was applied to approximately 15 similarly complex sentences throughout the report to improve readability for a broader audience, but all technical content, mathematical derivations, and scientific conclusions originated from our team’s analysis.

3. Google Gemini (accessed via Cursor IDE auto-completion, February 2026)

Purpose: Code auto-completion and syntax assistance during implementation of:

- The Two-Path Regression model with season/week fixed effects and contestant-clustered standard errors (Problem 3)
- Counterfactual elimination-chain simulation loops (Problems 2 and 4)
- Hit-and-Run MCMC sampler with adaptive step-size control (Problem 1)

Example auto-completion: When implementing panel data regression with clustered standard errors, Cursor suggested the correct `statsmodels` syntax:

```
import statsmodels.formula.api as smf
model_judge = smf.ols(
    'judges_total ~ C(season) + C(week) + pro_experience + ...',
    data=panel_df
).fit(cov_type='cluster', cov_kws={'groups': panel_df['celebrity']})
```

We verified this syntax against the official statsmodels documentation v0.14.0 and confirmed the correct implementation of within-group correlation adjustment.

How we used this: Auto-completion accelerated coding efficiency by approximately 30%, particularly for boilerplate data manipulation and visualization code. However, all modeling logic, algorithm design, parameter choices, and validation strategies were determined by our team through collaborative discussion. We manually reviewed and tested every code block, and all results were independently verified against theoretical expectations.

4. Additional Embedded AI Tools

- **Python Scientific Libraries:** We used standard implementations from NumPy (v1.24), SciPy (v1.11), Pandas (v2.0), Statsmodels (v0.14), and scikit-learn (v1.3). These libraries contain embedded optimization algorithms (e.g., `scipy.optimize.linprog` for feasible region checks, `scipy.stats.kendalltau` for rank correlation) and statistical estimators. We verified all outputs against theoretical expectations and cross-validated key results using alternative implementations.
- **Matplotlib/Seaborn:** All figures were generated using Matplotlib (v3.7) and Seaborn (v0.12) with standard plotting functions. Embedded auto-formatting features (e.g., automatic color scheme selection, layout optimization) were applied after manual verification for accessibility and visual clarity.
- **LaTeX Auto-formatting:** We used Overleaf's built-in syntax checking and auto-completion for mathematical notation and bibliography management. All mathematical derivations were independently verified by hand.

Verification and Team Responsibility

All AI-generated content—whether from direct queries, code auto-completion, or embedded algorithms—was carefully reviewed by our team for:

1. **Mathematical accuracy:** All equations, derivations, and algorithmic logic were independently verified against theoretical foundations and validated through numerical tests.

2. **Data integrity:** All data processing steps were checked for correctness through unit tests, edge-case validation, and cross-validation against alternative implementations.
3. **Citation accuracy:** We manually verified that all references cited in our report (including those suggested by AI tools) actually exist and are correctly attributed. We checked original sources to ensure we are not inadvertently plagiarizing.
4. **Scientific validity:** All modeling decisions, problem formulations, and conclusions were developed through collaborative team discussion drawing on domain knowledge in Bayesian inference, causal inference, optimization theory, and applied statistics. AI tools served as productivity aids for implementation efficiency and language refinement, but did not substitute for human creativity and critical thinking.

We acknowledge that AI tools have limitations including potential biases, hallucinations, and contextual misunderstandings. Our team takes full responsibility for the accuracy and integrity of all content in this report.